The Gate and the First Flow: Program ω, Stage C's Opening Move

Harald Rønning

Draft — open, not closed. Companion to the frozen ω-plan (Stage B closed on the two-sign question; this note picks up Stage C: the Gaussianity gate, the arithmetic fork it surfaced, and the first flow integration on two anchors). Every claim below is two-implemented unless explicitly marked otherwise; the error ledger is part of the record because the catches, this time more than any prior arc, are the substance of what was learned.

---

1. Charter

Stage B closed with $m(u)$'s asymptotic signs undetermined-without-dynamics — the missing datum handed forward to Stage C. This note covers Stage C's own opening: constructing the Gaussianity gate the frozen plan's P3 amendment specifies, the extended arithmetic fork that construction surfaced, and the first flow integration once the fork closed. It ends without a verdict, by design — the two anchors integrated here produced genuinely different, only partially interpreted behavior, and that openness is the honest state of the record, not a placeholder for a conclusion not yet written.

2. The Gaussianity gate: construction and its two-implementation

[FACT, two-implemented] The candidate dissipative dynamics is built from linear jump operators on the anchor's own real-linear operator L_A, expressed in a mode basis built from the gauge module's complex structure J_0. In quadratures $r = (x_1, p_1, ...)$, each jump $L_j = c_j^T r$, with $M = \sum_j c_j c_j^\dagger$, the covariance obeys the closed-form Lyapunov equation

$$\dot{\sigma} = A\sigma + \sigma A^T + D, \qquad A = -\Omega\,\mathrm{Im}(M), \quad D = \Omega\,\mathrm{Re}(M)\,\Omega^T,$$

σ centered and symmetrized, Ω the standard symplectic form. This formula was independently derived and cross-verified against direct Fock-space simulation (finite truncation, exact master-equation dissipator), converging to machine precision as truncation grows — the formula itself was never in dispute once both sides tested it against ground truth rather than each other.

3. The arithmetic fork, and what it actually was

[LEDGER, the arc's own record] What followed was not a physics disagreement but a five-layer convention stack, each layer catching a genuine error before the next could hide behind it.

Layer 1 — stale coordinates. The anchor L_A built fresh in this session's own basis was combined with a saved J_0 file from an earlier part of this long session, built on a silently different basis realization. Both objects passed every intrinsic check individually ($J_0^2 = -\mathrm{Id}$ exact; L_A's tunnel spectrum exact) — alignment between them was never tested until a full rebuild on consistent inputs was forced. Root cause named precisely: **saved-object coordinate provenance** — a file can carry validity without carrying the coordinate frame it was validated in, and two individually-correct objects from different moments of one session are not automatically combinable. Resolution: full rebuild from one `cd16` call, no saved files. Confirmed split: $\|C\|^2 = 4$, $\|D\|^2 = 12$, matching the independently-derived basis-free invariant ($\|L_c\|^2/2$, $\|L_a\|^2/2$ for the J_0-complex-linear/antilinear parts of L_A) exactly.

Layer 2 — the definitional fork. Two constructions for "the anchor's dynamics under the mirror orientation" were each built and verified exact: (a) the same quadrature machine fed the mirror's conjugated inputs, giving $A_2 = A_1$ (orientation-blind) and $[P, A_1] = 0$; (b) the literal adjoint jump-set $\{L_j^\dagger\}$, giving $A_2 = -A_1$ (arrow-duality, loss↔gain). Both are exact theorems about two different, well-defined objects — not competing answers to one question. **[THM, resolved as a duality, not a choice]** The two compose: orientation-flip followed by the construction equals time-reversal of the construction. Neither reading is privileged; a future document naming "the mirror candidate" states locally which half it means.

Layer 3 — a mislabeled scalar. A proposed "kernel diffusion $= N(A)$" identity was traced to its source and found to be a Frobenius norm of a 4×4 block, reported as a per-direction rate. Rerun with the value computed two independent ways (via the assembled D matrix, and directly as $\sum_j |c_j \cdot k|^2$) gives $k^T Dk = 1.0$ per kernel direction, both routes agreeing to machine precision. Bonus structural fact, confirmed: $\mathrm{Re}(M)$ annihilates the kernel directly — the jump vectors are exactly orthogonal to it — so the diffusion the kernel experiences arrives entirely through the Ω-dressing, from its symplectic partner directions, not from any direct coupling.

Species logged, this arc, both sides: *saved-object coordinate provenance* (mine); *output-attribution blur*, *rationalized disconfirmation*, *theorem-scope inflation*, *quantity-type mislabeling* (F's, each independently self-diagnosed and ledgered in turn). The countermeasure common to all: a cheap, mechanical check (print the actual matrices consumed, not re-derived ones) run *before* trusting an internally-consistent-looking result, not after a contradiction forces it.

4. Six results, closed

Confirmed two-implemented, on fully consistent inputs, no open threads:

1. **Split**: $\|C\|^2 = 4$, $\|D\|^2 = 12$, $\mathrm{tr}(A) = 8$.
2. **Drift spectrum**: $\{0^8, +1^8\}$ for the classic anchor $e_1 + e_{10}$.
3. D**-equality**: $D_2 = D_1$ under either reading of the orientation flip.
4. **Kernel drift-freeze**: exact to 10^{-16}, refined to $\mathrm{Re}(M)$-annihilation.
5. **The 84-diagonal survey**: exactly three classes, $12/60/12$, trace $\in \{+16, +8, -8\}$.
6. **The strut cross-tabulation**: the mediator kite is uniquely class-pure (all 12 diagonals in the middle class); every other kite splits identically $2+8+2$ across the three classes, verified via the seam note's own assessor-product strut rule.

5. The first flow: two anchors, three discriminators, open result

[EXPERIMENT, one-pass, not yet independently re-verified] Two anchors integrated from a shared quasi-free initial state ($\nu = 1.5$): the classic expanding-class anchor ($e_1 + e_{10}$, $\mathrm{tr}\, A = +8$) and a contracting-class anchor ($e_4 + e_9$, $\mathrm{tr}\, A = -8$, drift spectrum $\{-2^4, 0^{12}\}$ — a genuinely different structure from a simple sign-flip of the expanding case).

Discriminator 1 (deepens vs. wanders). The expanding anchor's χ_{kin} splits from the uniform initial value into two branches, both growing monotonically and without bound — coordinated growth, genuinely satisfying Becoming's partial-order criterion for depth-motion, even as an unbounded runaway. The contracting anchor's χ_{kin} splits into *three* branches instead: one decays toward the vacuum value and settles there, one stays exactly frozen, one grows linearly without bound — simultaneously shrinking, static, and growing. This is wandering, not deepening, by the same criterion.

Discriminator 3 (family-tracking). Neither anchor stays on the quasi-free line; both depart it immediately upon integration. The rates differ by roughly two orders of magnitude — the expanding anchor's departure explodes, the contracting anchor's departure is comparatively gentle.

What this note does not do: resolve which of these — if either — constitutes "settling" in the corpus's sense. The anchor named for stability (by its drift matrix) is the one that wanders in χ_{kin}; the anchor that deepens cleanly is the one that never approaches any stationary point. Both facts are genuine and neither was anticipated by either anchor's own classification. **The open ending is deliberate**: this is a first pass, one integration each, not yet cross-checked by a second implementation, and the interpretive question — what a mixed shrink/static/grow spectrum *means* against the corpus's own settling doctrine — is not yet attempted here.

6. Open items, carried forward

- Independent re-verification of §5's flow integration (a second implementation of the ODE solve and the discriminator computations).
- Discriminator 2 (direction structure against the tunnel bands) not yet run on either anchor's actual trajectory.
- Whether the contracting anchor's three-way split (decay / frozen / linear growth) has a structural explanation tied to its $\{-2^4, 0^{12}\}$ spectrum, the way the expanding anchor's coordinated growth ties to its $\{0^8, +1^8\}$ spectrum.
- The registered guess on the strut cross-tabulation's own $2/8/2$ pattern (the extreme pairs tied to the strut's J_0-partner structure) remains a guess, not computed.
- Whether either anchor's behavior, or some structural combination, should inform REG-2's successor question (the anchor-supplier problem) remains entirely untouched by this note.

Created in collaboration with Claude (Anthropic). Draft, open ending as stated in §5; all conclusions and any errors remain the author's responsibility.